

07

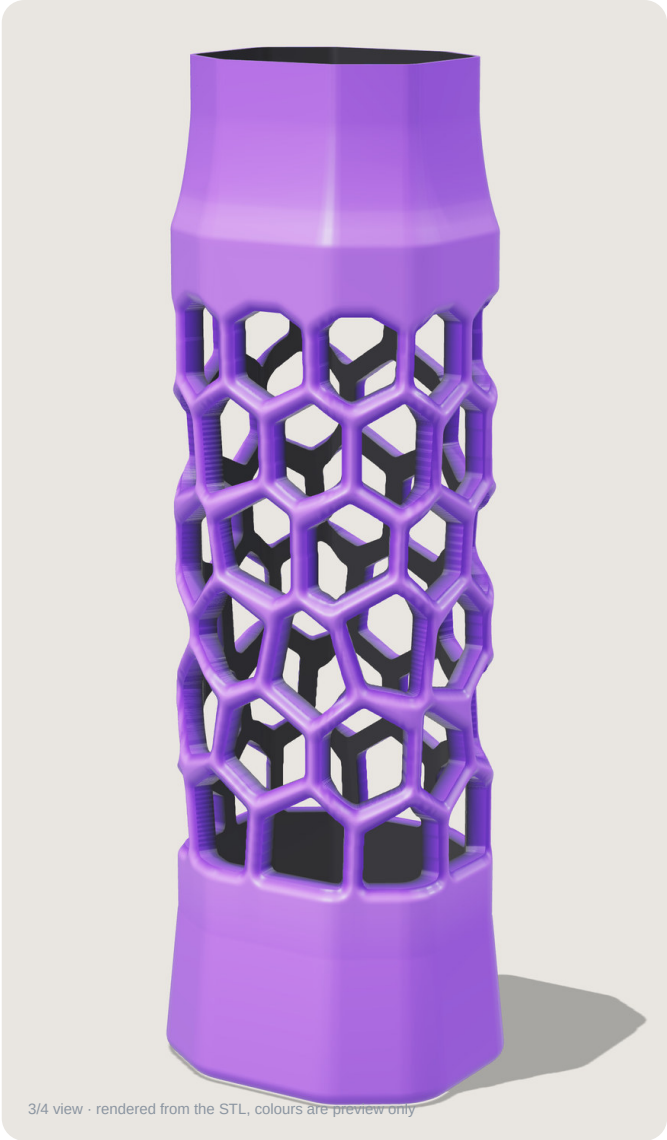
VORONOI-CORE

Voronoi rib lattice, windows cut through the wall for weight saving and airflow.

40.2 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	39.1 × 33.0 mm
Peak texture height	3.50 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	32.41 cm³ / 259 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		40.2 g
PETG	1.27 g/cm³		41.2 g
TPU	1.21 g/cm³		39.2 g
ABS	1.04 g/cm³		33.7 g

50 g limit

COLOURWAY



#8A54E4

3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 13.5 m (100 % flow)

DESIGN INTENT

Irregular Voronoi cells become 2.6 mm ribs with rounded crests, and the cells themselves are open windows through the wall – so the grip is ventilated and the lightest of the ten. The lattice is relaxed for even cells but keeps a natural irregularity.

Window roofs are trimmed; the longest bridge is 3.8 mm.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical; 54 windows open the wall
Outer envelope vs reference	+0.003 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	3.8 mm

KEY FEATURES

- 54 through-windows in the band z 27.5 – 103.5 mm
- 2.6 mm ribs standing at the full 3.5 mm crest
- Window roofs trimmed to 60 degrees, longest bridge 3.8 mm

FILES

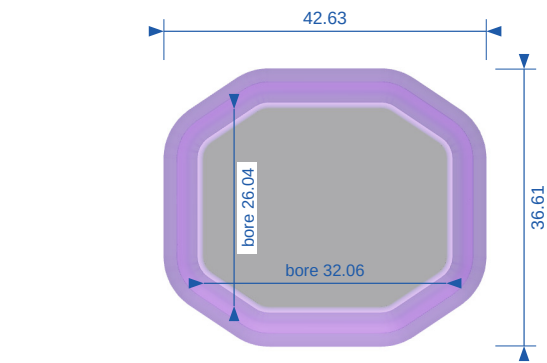
stl/07_voronoi_core.stl	9.7 MB · 194,162 triangles
3mf/07_voronoi_core.3mf	2.3 MB · colour

✓ FITMENT VERIFIED

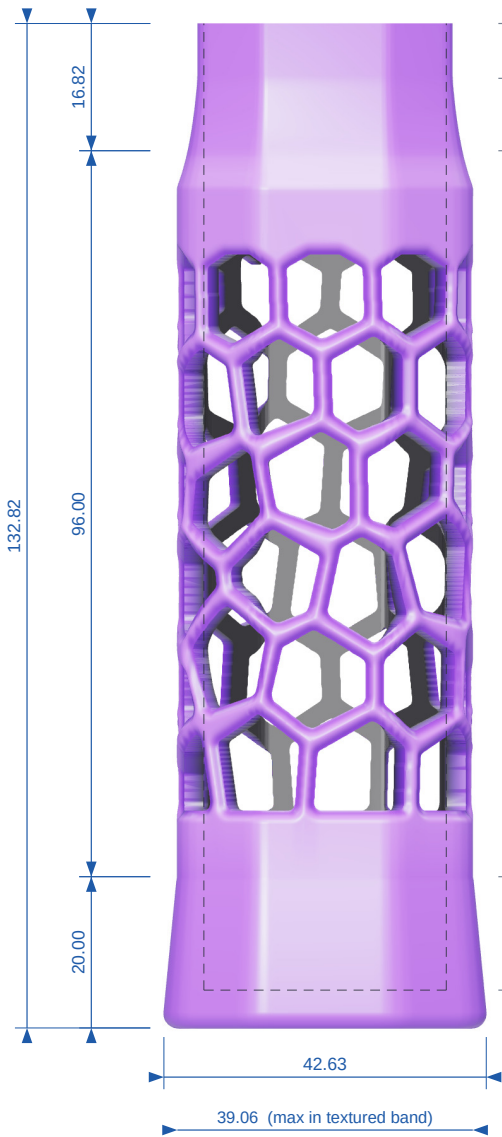
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

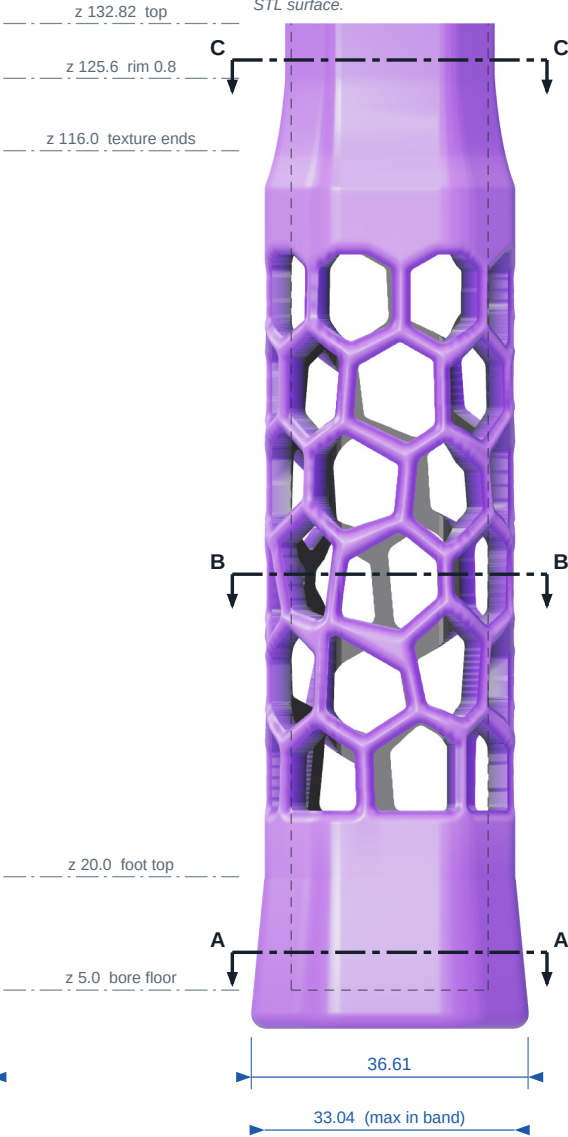
✓ 3MF VALIDATED



TOP VIEW (from +Z)



FRONT VIEW (from -Y)



RIGHT VIEW (from +X)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	39.06 × 33.04
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.50
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.

SCALE 1 : 1



PART

VORONOI-CORE

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

MASS (PLA)

40.2 g

REVISION

2

UNITS

millimetres

INFILL

100 %

VOLUME

32.41 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

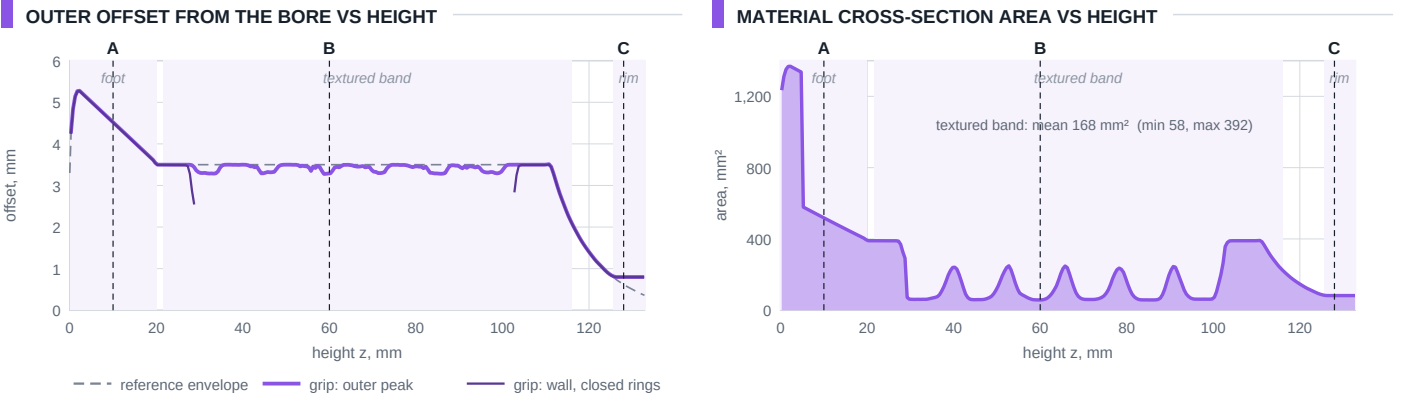
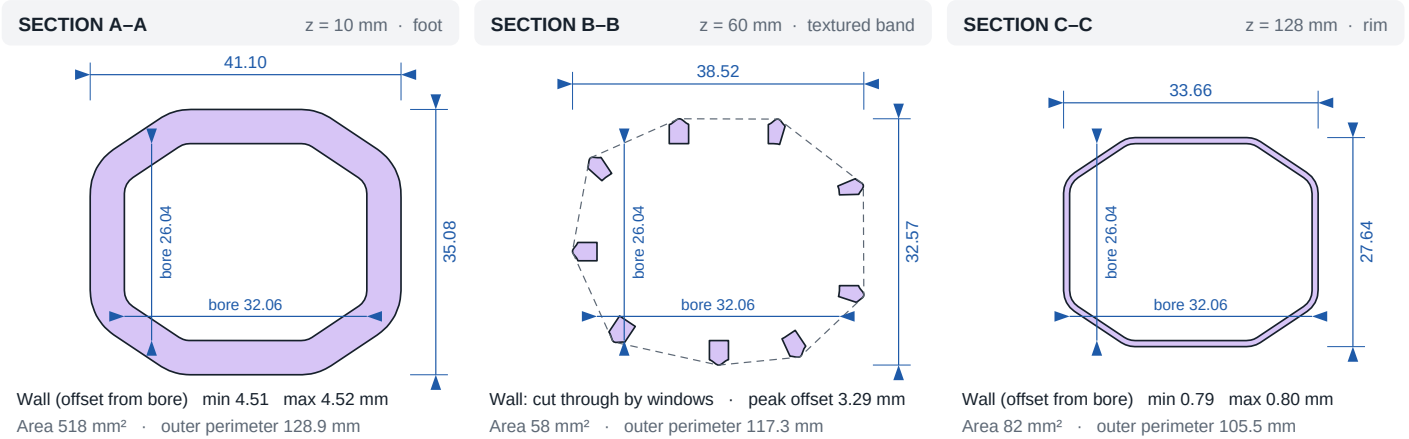
259 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	39.06 × 33.04 mm
Circumference in band	116.3 – 122.5 mm
Volume / surface area	32.41 cm ³ / 259.0 cm ²

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / windows
Rim wall (design / measured min)			0.80 / 0.80 mm
Peak texture height above bore			3.50 mm

OUTER DIMENSIONS BY HEIGHT

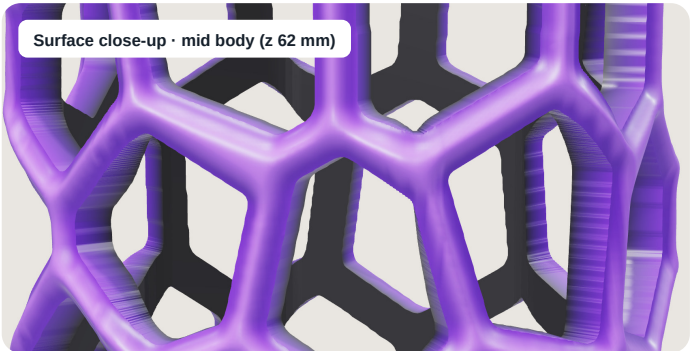
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	38.63	32.65	116.8	62	open	3.31
45.0	38.57	32.62	117.2	59	open	3.30
60.0	38.52	32.57	117.3	58	open	3.29
75.0	38.81	32.61	117.5	88	open	3.50
90.0	38.91	32.86	120.5	212	open	3.46
105.0	39.06	33.04	122.5	390	3.49	3.50
116.0	36.27	30.24	113.7	225	2.10	2.10
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

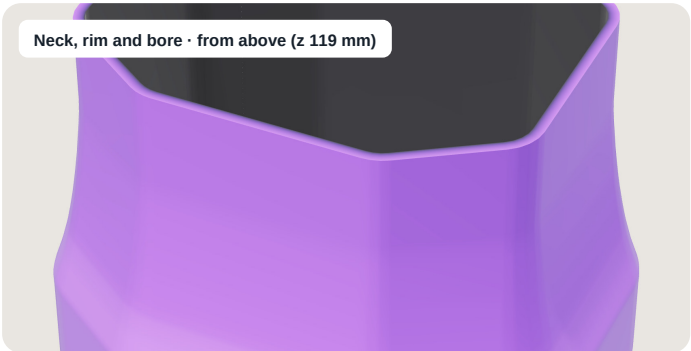
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only

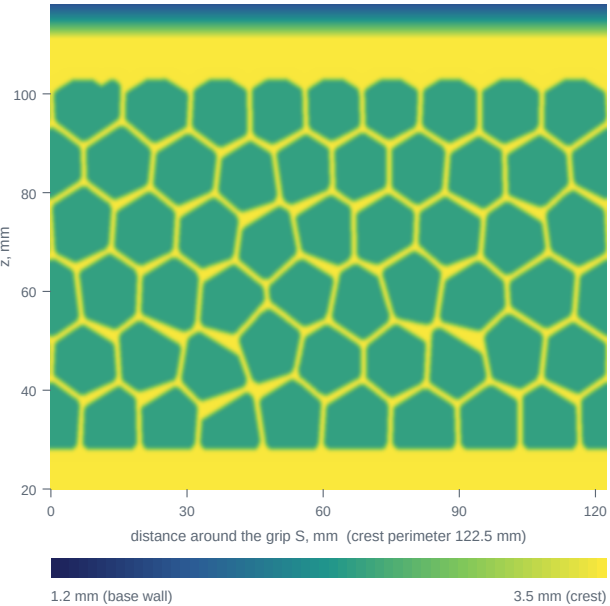


Surface close-up · mid body (z 62 mm)



Neck, rim and bore · from above (z 119 mm)

UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Windows	54, through the wall
Rib width	2.6 mm
Rib height	3.50 mm above bore
Windowed band	z 27.5 – 103.5 mm (76 mm)
Window corner radius	1.0 mm
Roof limit	60 deg overhang, 4 mm flat
Open area	about 64% of the band

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 54 (54 windows)
Bore vs reference	identical (windows open wall)
Envelope over reference	+0.003 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	3.8 mm